Accumulation points for the Collatz conjugacy map in $\mathbb{R}$

Abstract

The Collatz conjecture splits into two independent assertions: that no trajectory diverges, and that the only cycle is the trivial one. Lagarias' periodicity conjecture is the divergence half, extended from the integers to the rationals with odd denominator, and by the Bernstein–Lagarias conjugacy it becomes a question about which aperiodic words can be parity vectors. López and Stoll transported this question to $\mathbb{R}$, where a word with an accumulation point in its forced trajectory is excluded as a witness; for their Example 6 they could not decide whether such a point exists. We prove that it does. The argument rests on the identity $3^{\log 2/\log 3} = 2$, which makes every discontinuity of a certain limit function an upward jump and so produces an honest root of a level set; the only computational input is a pair of inequalities verified in exact rational interval arithmetic. We then embed the example in a one-parameter family, determine exactly for which parameters one distinguished fibre is excluded, identify the limit of the family in closed form, and isolate a scalar R_n whose boundedness governs recurrence throughout the class. A counterexample shows that recurrence is *not* governed by the obvious candidates. We are explicit about what this does not do: it does not bear on either half of the Collatz conjecture, for reasons we set out precisely.

1 Introduction

The two halves

Let T be the $3x+1$ map, $T(x) = x/2$ for x even and $T(x) = (3x+1)/2$ for x odd. The Collatz conjecture states that the trajectory of every positive integer reaches the cycle $1 \mapsto 2 \mapsto 1$. It is the conjunction of two logically independent assertions:

(D) *no divergence*: every trajectory is bounded, hence eventually cyclic;

(C) *no nontrivial cycle*: the only cycle among the positive integers is $\{1, 2\}$.

Neither implies the other, and they are attacked by quite different means — (C) by lower bounds on the length of a hypothetical cycle, (D) by growth and density estimates.

The map T extends to the 2-adic integers $\mathbb{Z}_2$, and there it acts on the ring $\mathbb{Q}_{\text{odd}}$ of rationals with odd denominator, which is exactly the set of $x \in \mathbb{Z}_2$ whose orbit could be rational. Lagarias' *periodicity conjecture* asserts that the trajectory of every $x \in \mathbb{Q}_{\text{odd}}$ is eventually cyclic. Restricted to the positive integers this is precisely (D); so the periodicity conjecture *is* the divergence half of Collatz, posed over a larger ring. It says nothing about (C): eventual cyclicity does not assert which cycles occur. Thus

$$\text{periodicity conjecture} \implies \text{(D)}, \qquad \text{Collatz} = \text{(D)} \wedge \text{(C)},$$

and a proof of the periodicity conjecture would settle the harder-looking half of Collatz while leaving the cycle half untouched.

Words

Bernstein and Lagarias [1] showed that the parity map $x \mapsto v(x) = (T^k(x) \bmod 2)_{k\geq 0}$ is a homeomorphism of $\mathbb{Z}_2$ conjugating T to the shift, with inverse

$$\Phi\Big(\sum_{i\geq 0} 2^{d_i}\Big) = -\sum_{i\geq 0} \frac{2^{d_i}}{3^{i+1}}, \qquad d_0 < d_1 < \cdots . \tag{1}$$

Under this dictionary a trajectory is eventually cyclic exactly when its parity vector is eventually periodic, so the periodicity conjecture becomes: *no aperiodic word is the parity vector of an element of* $\mathbb{Q}_{\text{odd}}$. A counterexample would be a single aperiodic word v with $\Phi(v) \in \mathbb{Q}_{\text{odd}}$. One is therefore led to examine aperiodic words one at a time and try to rule each out.

Two facts delimit the search. First [2, Theorem 1]: if some $\zeta \in \mathbb{Q}_{\text{odd}}$ has a divergent trajectory, then necessarily $\liminf_\ell h(\ell)/\ell = \log 2/\log 3$, where $h(\ell)$ counts the ones among the first ℓ digits of its parity vector. So a counterexample can only have this one critical density, and every word considered below has it. Second [2, p. 3]: an infinite subset of $\mathbb{Q}_{\text{odd}}$ has no accumulation point in $\mathbb{R}$; hence exhibiting one accumulation point in the orbit of ζ proves $\zeta \notin \mathbb{Q}_{\text{odd}}$ and eliminates that word.

The real picture

López and Stoll [2] transported this to $\mathbb{R}$: for a word v one reads (1) as a real series, obtaining $\Phi_{\mathbb{R}}(v)$ when it converges, and lets T act on $\mathbb{R}$ through the branches prescribed by v, irrespective of parity. Writing σ for the shift, the *forced trajectory* of $\zeta = \Phi_{\mathbb{R}}(v)$ has terms $\Phi_{\mathbb{R}}(\sigma^t v)$. They eliminate every word of supercritical density, and among critical-density words they settle several; for their Example 6 the question was left open:

> "Example 6 is such a case where we cannot guarantee a (probably existing) accumulation point." [2, p. 4]

Our main result settles this case.

Theorem 1. *Let v be the word of [2, §14] and $\zeta = \Phi_{\mathbb{R}}(v)$. Then ζ is an accumulation point of its own forced trajectory.*

Sections 4–6 develop the surrounding structure: the example lies in a family indexed by a parameter k, for which the whole apparatus is parameter-free (Proposition 11); one distinguished fibre of the reduction is excluded for exactly $k \geq 3$ (Theorem 16); the family has an explicit limit (Proposition 17); and a single scalar governs recurrence throughout the class (Theorem 18), though not the one the examples suggest (Remark 20).

Where this sits

It is worth being blunt about the distance to the conjectures. The programme sketched above would, if carried out for *every* aperiodic word, prove the periodicity conjecture and hence half (D) of Collatz; half (C) would remain untouched by any of it. What is proved here is one step of that programme — a single word, and then a family — together with two structural facts about how far the step generalises.

Both facts are negative, and both are worth recording. The associated word of [2, Example 7] has the same critical density and provably *no* accumulation point, so the method cannot be universal: it will never eliminate all aperiodic words. And Corollary 19

constrains the *real* value $\Phi_{\mathbb{R}}(v)$, whereas the periodicity conjecture concerns the 2-*adic* value $\Phi(v)$; these are limits of the same rational partial sums in different completions, and we cannot show they agree. Section 7 locates that missing implication exactly, as a growth estimate (8) for the orbit of the hypothetical rational, which is the sharpest form in which we can state what is absent. Nothing below should be read as progress on the Collatz conjecture itself.

2 Notation

Throughout $\alpha = \log 2/\log 3$ and $\gamma = \alpha/(1-\alpha) = \log 2/\log(3/2)$, so that

$$3^{\alpha} = 2, \qquad \gamma \log \tfrac{3}{2} = \log 2. \tag{2}$$

Every constant below is a consequence of (2). We write $\{x\}$ for the fractional part.

Fix $k \geq 2$ and set $f(n) = \lceil n/k \rceil$, $g(n) = n - f(n)$. The word is

$$v = \prod_{n \geq 1} 1^{m_n} 0^{f(n)}, \qquad m_n = \left\lfloor \frac{(L_{n-1}+n)\alpha - H_{n-1}}{1-\alpha} \right\rfloor, \tag{3}$$

where L_n, H_n are the length and the number of ones of the first n factors, $L_0 = H_0 = 0$; thus $L_n = L_{n-1} + m_n + f(n)$ and $H_n = H_{n-1} + m_n$. Example 6 of [2] is $k = 6$; its first values are $m_1, \ldots, m_8 = 1, 4, 3, 3, 4, 3, 4, 5$. Let Z_n be the number of zeros among the first n factors and $G(n) = Z_{n-1} + n$; from $f(n) = \lceil n/k \rceil$ one computes that $G(kM + s)$ is a quadratic polynomial in M with leading coefficient $k/2$.

Put $D_n = \alpha L_n - H_n$. Let $z_n = \Phi_{\mathbb{R}}(\sigma^{L_n} v)$ be the value of the forced trajectory at the end of the n-th factor, so $z_0 = \zeta$.

Lemma 2. *$H_n = \lfloor \gamma G(n) \rfloor$ and $D_n = -\alpha g(n) + \beta_n$ with $\beta_n = (1-\alpha)\{\gamma G(n)\} \in [0, 1-\alpha)$. Consequently*

$$D_{n+r} - D_n \leq -\alpha \left\lfloor \tfrac{(k-1)r}{k} \right\rfloor + (1-\alpha), \qquad D_n \leq (1-\alpha) - \tfrac{(k-1)\alpha}{k}(n-1). \tag{4}$$

Proof. As $H_{n-1} \in \mathbb{Z}$ it may be absorbed into the floor in (3), giving

$$H_n = \left\lfloor \frac{\alpha L_{n-1} + n\alpha - \alpha H_{n-1}}{1-\alpha} \right\rfloor = \lfloor \gamma (L_{n-1} - H_{n-1} + n) \rfloor = \lfloor \gamma G(n) \rfloor.$$

Then $D_n = \alpha Z_n - (1-\alpha) H_n = \alpha(Z_n - G(n)) + \beta_n$ and $Z_n - G(n) = f(n) - n = -g(n)$. The estimates follow from $\beta_{n+r} - \beta_n < 1 - \alpha$ and $g(n) = \lfloor (k-1)n/k \rfloor$. □

3 Proof of Theorem 1

Throughout this section $k = 6$. Let $d_0 < d_1 < \cdots$ be the positions of the ones of v.

Lemma 3. $z_n = -\displaystyle\sum_{r \geq 0} 3^{\,D_{n+r} - D_n} \left(1 - (2/3)^{m_{n+r+1}}\right)$.

Proof. The ones of $\sigma^{L_n} v$ are the $d_{H_n+i} - L_n$, so $z_n = -\sum_{i \geq 0} 2^{\,d_{H_n+i} - L_n} 3^{-(i+1)}$. The ones of the factor $n+r+1$ occupy the positions $L_{n+r}, \ldots, L_{n+r} + m_{n+r+1} - 1$ and carry the indices $H_{n+r}, \ldots, H_{n+r} + m_{n+r+1} - 1$, so they contribute $-2^{\,L_{n+r} - L_n} 3^{-(H_{n+r} - H_n)} \cdot \frac{1}{3} \sum_{i < m} (2/3)^i$ with $m = m_{n+r+1}$. The inner sum is $3(1-(2/3)^m)$, and $2^{\Delta L} 3^{-\Delta H} = 3^{\,\alpha \Delta L - \Delta H} = 3^{\,D_{n+r} - D_n}$ by (2). □

Let $w_n = -\sum_{r\ge 0} 3^{\,D_{n+r}-D_n}$ be the corresponding sum with all damping factors removed.

Proposition 4. *$0 \le z_n - w_n \le 9\rho^{\,n+1}$ with $\rho = 2^{-1/6}$.*

Proof. By (4) and (2), $3^{\,D_{n+r}-D_n} \le 3^{1-\alpha}3^{\alpha}q_1^r = 3q_1^r$ where

$$q_1 = 3^{-5\alpha/6} = 2^{-5/6}. \tag{5}$$

From (3), $m_n \ge \gamma(n+5)/6-1$, and $(2/3)^{\gamma/6} = 2^{-1/6} = \rho$ by (2), so $(2/3)^{m_{n+r+1}} \le \frac{3}{2}\rho^{\,n+r+1}$. The r-th term of $z_n - w_n = \sum_r 3^{\,D_{n+r}-D_n}(2/3)^{m_{n+r+1}}$ is therefore at most $\frac{9}{2}q_1^r\rho^{\,n+r+1}$. Now

$$q_1\rho = 2^{-5/6}\cdot 2^{-1/6} = \tfrac{1}{2},$$

so the series is geometric of ratio $\frac{1}{2}$ and sums to at most $9\rho^{\,n+1}$. □

The coincidence $q_1\rho = \frac{1}{2}$ is not accidental: by (2) the loss of drift and the gain of damping per factor are reciprocal at the critical density, for every k.

Write $n = 6M + s$ with $0 \le s \le 5$, and set $u = \{\gamma G(n)\}$, $v = \{\gamma M\}$ (no confusion with the word should arise). Since $6M \equiv 0 \pmod 6$ we have $f(n+r) - f(n) = \lceil (s+r)/6\rceil - \lceil s/6\rceil$, and expanding the quadratic G gives $G(n+r) - G(n) = rM + C_s(r)$ with $C_s(r) = G(s+r) - G(s)$ free of M. As $r\lfloor\gamma M\rfloor \in \mathbb{Z}$, $\{\gamma G(n+r)\} = \{u + rv + \gamma C_s(r)\}$. With $A_s(r) = \alpha\big((\lceil (s+r)/6\rceil - \lceil s/6\rceil) - r\big)$, Lemma 2 therefore yields

$$w_{6M+s} = F_s(u,v), \qquad F_s(u,v) = -\sum_{r\ge 0} 3^{\,A_s(r)+(1-\alpha)(\{u+rv+\gamma C_s(r)\}-u)}. \tag{6}$$

Thus w is a fixed function of two variables sampled along an orbit. Since $A_s(r) \le \alpha(1-5r/6)$ and the second exponent is less than $1-\alpha$, every term of (6) is smaller than $3q_1^r$; the series converges uniformly and F_s has one-sided limits everywhere. Moreover $\{\cdot\}$ is right-continuous, hence so is F_s.

Lemma 5. *Every discontinuity of F_s is an upward jump.*

Proof. As a phase $u+rv+\gamma C_s(r)$ crosses an integer upward, $\{\cdot\}$ drops by 1, so the exponent of that term drops by $1-\alpha$ and the term is multiplied by $3^{\alpha-1} = 2/3$ by (2). Every term is positive, so each jump of $F_s = -\sum$ is upward; uniform convergence transfers this to the sum. □

Lemma 6. *Let $f : [a,b] \to \mathbb{R}$ have one-sided limits everywhere, be right-continuous, and satisfy $f(x^-) \le f(x)$ for all x. If $f(a) > c > f(b)$ then $x^* = \sup\{x : f(x) > c\}$ satisfies $f(x^*) = c$, and f has no jump at x^*.*

Proof. The set $A = \{x : f(x) > c\}$ contains a. If $x^* = b$ then $x_j \uparrow b$ with $f(x_j) > c$ would give $f(b) \ge f(b^-) \ge c$, contradicting $f(b) < c$; so $x^* < b$. No $x > x^*$ lies in A, so $f \le c$ on $(x^*, b]$ and right-continuity gives $f(x^*) \le c$. Points of A increasing to x^* give $f(x^{*-}) \ge c$, and $f(x^*) \ge f(x^{*-})$. Hence $c \le f(x^{*-}) \le f(x^*) \le c$. □

Lemma 7. *If $F_s(u_0,\cdot)$ has no jump at v^* and $u_0 \in (0,1)$, then F_s is continuous at (u_0, v^*) as a function on $\mathbb{T}^2$.*

Proof. The jump of $F_s(u_0,\cdot)$ at v^* is the sum of the jumps of the individual terms, all of one sign by Lemma 5; a sum of nonnegative terms vanishes only if each does. For $r \ge 1$ the phase is strictly monotone in v, so no phase line $u + rv + \gamma C_s(r) \in \mathbb{Z}$ passes through (u_0, v^*); the term $r = 0$ is constant. Now given $\varepsilon > 0$ choose R with $3q_1^{R+1}/(1-q_1) < \varepsilon/3$. The tail contributes less than $\varepsilon/3$ at every point of $\mathbb{T}^2$, while the head is a finite sum whose discontinuity set is a finite union of lines, none through (u_0, v^*); it therefore varies by less than $\varepsilon/3$ on a ball there. □

The care in Lemma 7 is necessary: the full discontinuity set is a countable union of lines of spacing $1/r$, hence dense, and no jump-free neighbourhood exists.

Lemma 8. *For each s, the sequence $\big(\{\gamma G(6M+s)\},\{\gamma M\}\big)_{M\ge 0}$ is equidistributed in $\mathbb{T}^2$.*

Proof. γ is irrational, since $\gamma = p/q$ would give $2^{q+p} = 3^p$. For $(h_1,h_2) \neq (0,0)$ the polynomial $h_1\gamma G(6M+s) + h_2\gamma M$ has an irrational non-constant coefficient, namely $3h_1\gamma$ if $h_1 \neq 0$ and $h_2\gamma$ otherwise. Apply Weyl's theorem and the Weyl criterion on $\mathbb{T}^2$. □

Proposition 9. $F_3\big(\frac{37}{100},\frac{369}{4000}\big) > \zeta > F_3\big(\frac{37}{100},\frac{37}{400}\big)$.

This is a finite computation, carried out in interval arithmetic whose endpoints are exact rationals: $\log 2$ and $\log 3$ come from the artanh series with its explicit remainder, 3^x from an argument-reduced exponential with the Lagrange remainder, ζ from its defining series with the geometric tail of (4), and F_3 from (6) truncated at $r = 150$ with the tail bound $3q_1^{151}/(1-q_1) < 9.1\cdot 10^{-38}$. Each floor and fractional part occurring is certified not to straddle an integer. The certified values are

$$\begin{aligned}
\zeta &= -2.29230550205470085073240735 57\ldots\\
F_3(\tfrac{37}{100},\tfrac{369}{4000}) &= -2.29226313663525243593951225 53\ldots\\
F_3(\tfrac{37}{100},\tfrac{37}{400}) &= -2.29256379643281597797707427 33\ldots
\end{aligned}$$

with margins $4.2\cdot 10^{-5}$ and $2.6\cdot 10^{-4}$ against arithmetic widths below 10^{-37}.

Proof of Theorem 1. Take $s = 3$ and $u_0 = 37/100$. By Proposition 9 and Lemmas 5, 6, the function $F_3(u_0,\cdot)$ meets the level ζ at $v^* = \sup\{v : F_3(u_0,v) > \zeta\}$ without a jump there, so $F_3(u_0,v^*) = \zeta$ and, by Lemma 7, F_3 is continuous at (u_0,v^*). By Lemma 8 there are $M_1 < M_2 < \cdots$ with $(\{\gamma G(6M_j+3)\},\{\gamma M_j\}) \to (u_0,v^*)$, whence $w_{6M_j+3} = F_3(\cdot) \to \zeta$ by (6), and $z_{6M_j+3} \to \zeta$ by Proposition 4. □

Remark 10. The conclusion is a genuine one about the trajectory, not only about the auxiliary sequence: computing z_{6M+3} directly from Lemma 3 at orbit points approaching (u_0,v^*) gives $|z_{6M+3}-\zeta|$ proportional to the distance, with ratio between 0.88 and 1.20 at the five closest points with $M < 2\cdot 10^7$ (the nearest being $M = 4\,153\,063$ at distance $9.4\cdot 10^{-5}$).

4 A family

Nothing in §3 used the value 6 except through f. Keep (3) with $f(n) = \lceil n/k\rceil$ and write ζ_k, $F_s^{(k)}$ for the resulting objects.

Proposition 11. *For every $k \ge 2$, Lemmas 2–8 and Proposition 4 hold with*

$$q_1 = 3^{-\alpha(k-1)/k} = 2^{-(k-1)/k}, \qquad \rho = (2/3)^{\gamma/k} = 2^{-1/k}, \qquad q_1\rho = \tfrac{1}{2}.$$

The ratios $3^{\alpha-1} = 2/3$, $3^{1-\alpha} = 3/2$ and $3^{-\alpha} = 1/2$ do not involve k, so Lemma 5 is unchanged; $C_s(r)$ is free of M; and $G(kM+s)$ is quadratic in M with leading coefficient $k/2 \neq 0$.

Consequently the proof of Theorem 1 applies verbatim once a level crossing is exhibited.

Theorem 12. *Let $k \ge 2$. If $F_s^{(k)}(u_0,v_1) > \zeta_k > F_s^{(k)}(u_0,v_2)$ for some s and $v_1 < v_2$, then ζ_k is an accumulation point of its own forced trajectory.*

The hypothesis has been certified, by the computation described after Proposition 9, in the five cases of Table 1.

k	s	u_0	$[v_1, v_2]$	ζ_k
6	3	37/100	$[\frac{369}{4000}, \frac{37}{400}]$	$-2.29230550205470085\ldots$
4	2	1/60	$[\frac{257}{400}, \frac{129}{200}]$	$-2.43242829803240526\ldots$
5	1	7/40	$[\frac{9}{25}, \frac{73}{200}]$	$-2.34442124302516940\ldots$
8	3	1/40	$[\frac{4}{25}, \frac{33}{200}]$	$-2.26316625669379821\ldots$
2	0	19/24	$[\frac{47}{75}, \frac{377}{600}]$	$-3.51194122554282436\ldots$

Table 1: Certified crossings. Margins exceed $4 \cdot 10^{-5}$ throughout, against arithmetic widths below 10^{-21}.

5 The fibre $s = 0$

For $k \geq 3$ the crossings above occur only for $s \geq 1$, and this is not an accident of the search. Write $\lambda = 3^{1-\alpha} = 3/2$ and $a_r = 3^{A_0(r)} = 2^{\lceil r/k \rceil - r}$, so that $-F_0^{(k)}(u, v) = 1 + \sum_{r \geq 1} a_r \lambda^{x_r - u}$ with x_r the relevant fractional part.

Proposition 13. *$-F_0^{(k)}(u, v) > B_k$ for all (u, v), where*

$$B_k = 1 + \tfrac{2}{3} \sum_{r \geq 1} 2^{\lceil r/k \rceil - r} = 1 + \frac{4}{3} \cdot \frac{1 - 2^{-k}}{1 - 2^{-(k-1)}}.$$

B_k decreases strictly to $7/3$, so $B_k > 7/3$ for every k.

Proof. The term $r = 0$ equals 1; for $r \geq 1$ the factor $\lambda^{x_r - u}$ exceeds $\lambda^{-1} = \frac{2}{3}$, and $3^{A_0(r)} = 2^{\lceil r/k \rceil - r}$ by (2). Writing $r = (b-1)k + i$ with $1 \leq i \leq k$ gives $\sum_{r \geq 1} 2^{\lceil r/k \rceil - r} = (1 - 2^{-k}) \cdot 2/(1 - 2^{-(k-1)})$. Finally $x \mapsto (1 - x)/(1 - 2x)$ is increasing and $x = 2^{-k}$ decreases. □

B_k suffices for $k \geq 5$ but not below, and the deficiency is repaired by observing that the first two terms are coupled: $C_0(1) = 1$ and $C_0(2) = 3$ for every k, so with $x = \{u + v + \gamma\}$ one has $x_2 = \{2x + \gamma - u\}$.

Proposition 14. $\inf_{u,v} \left(\lambda^{x-u} + \frac{1}{2} \lambda^{x_2 - u} \right) = \dfrac{1}{\sqrt{2}} + \dfrac{1}{3}$, *whence $-F_0^{(k)} > B_k^\sharp := B_k + \frac{1}{\sqrt{2}} - \frac{2}{3}$.*

Proof. Put $e = x - u$ and $m = \lfloor 2e + u + \gamma \rfloor \in \{0, 1, 2, 3\}$, so that $x_2 - u = 2e + \gamma - m$ and the quantity is $\lambda^e + \frac{1}{2} \lambda^{2e + \gamma - m}$, strictly increasing in e. On the branch m the constraint $m \leq 2e + u + \gamma$ with $u < 1$ gives $e > (m - 1 - \gamma)/2$, and $x \geq 0$ gives $e > -1$. Evaluating at these endpoints gives $1.1112\ldots$, $\frac{1}{\sqrt{2}} + \frac{1}{3}$, $1.1993\ldots$, $1.3936\ldots$ for $m = 0, 1, 2, 3$. The minimum is at $m = 1$, $e = -\gamma/2$, where $\lambda^{-\gamma/2} = (3/2)^{-\gamma/2} = 2^{-1/2}$ by (2) and $\frac{1}{2} \lambda^{-1} = \frac{1}{3}$. The two terms were assigned $\frac{2}{3}(1 + \frac{1}{2}) = 1$ in B_k. □

For $k = 3$ even $B_3^\sharp$ falls short, and one uses all of the coupling.

Proposition 15. *Let $\theta_r = rv + \gamma C_0(r)$ and $Q_R = \inf_v \sum_{r \leq R} a_r \lambda^{\{\theta_r\}}$. Then $-F_0^{(k)} \geq B_k + \frac{2}{3}\big(Q_R - \sum_{r \leq R} a_r\big)$, and Q_R is the minimum of the finitely many values at $v = (n - \gamma C_0(r'))/r'$, $1 \leq r' \leq R$, $0 \leq n < r'$.*

Proof. Since $x_r - u = \{\theta_r\} - [\, u \geq 1 - \{\theta_r\}\,]$ and $\lambda > 1$, each term is at least $\frac{2}{3} a_r \lambda^{\{\theta_r\}}$ for every u; bounding the terms $r > R$ by $\frac{2}{3} a_r$ gives the inequality. On a cell where all $\lfloor \theta_r \rfloor$ are constant, $\{\theta_r\}$ is linear in v with slope $r > 0$, so every summand increases and the infimum over the cell is at its left endpoint. These are the stated points, taken modulo 1, the configuration being 1-periodic. At such a point $\theta_r = p + \gamma c$ with p, c rational, and as γ is irrational this is an integer only when $c = 0$ and $p \in \mathbb{Z}$, a case decided exactly. □

Theorem 16. *For $k \geq 3$ one has $F_0^{(k)} < \zeta_k$ everywhere, so the hypothesis of Theorem 12 cannot hold with $s = 0$. For $k = 2$ it does hold with $s = 0$ (Table 1).*

Proof. For $k \geq 7$: the words for different k agree on their first k factors, since $\lceil n/k \rceil = 1$ for $n \leq k$; so the sum of the first seven terms of $|\zeta_k| = \sum_n 3^{D_{n-1}}(1 - (2/3)^{m_n})$ is independent of k, and (4) bounds the remainder by $\frac{3}{2}q_1^6/(1-q_1)$ with $q_1 \leq 2^{-6/7}$. This gives $|\zeta_k| \leq 2.3231 < \frac{7}{3} < B_k$. For $k = 6, 5$ one has $|\zeta_6| = 2.29231 < B_6 = 2.35484$ and $|\zeta_5| = 2.34442 < B_5 = 2.37778$; note $|\zeta_5| > 7/3$, which is why B_k rather than its limit is used. For $k = 4$, $B_4 = 17/7$ is too small but $B_4^\sharp = \frac{17}{7} + \frac{1}{\sqrt{2}} - \frac{2}{3} = 2.46901 > 2.43243 = |\zeta_4|$. For $k = 3$, Proposition 15 with $R = 8$ gives $-F_0^{(3)} \geq 2.69267 > 2.66441 = |\zeta_3|$. □

At $k = 2$ every version of the bound fails, as it must; running the $R = 8$ estimate there returns $3.189 < 3.512 = |\zeta_2|$ and correctly declines to conclude.

Proposition 17. *As $k \to \infty$ the words agree on ever longer initial segments and converge to the word with one zero per factor and $H_n = \lfloor \gamma(2n-1) \rfloor$, $H_0 = 0$, $L_n = H_n + n$. Correspondingly $\zeta_k \to \zeta_\infty$ with*

$$|\zeta_\infty| = 1 + \frac{1}{2}\sum_{n \geq 1} 2^n \Big(\frac{2}{3}\Big)^{\lfloor \gamma(2n-1) \rfloor} = 2.2548974992816421641\ldots$$

Proof. With $f \equiv 1$ one has $Z_n = n$ and $G(n) = 2n - 1$, so Lemma 2 gives H_n and L_n. The n-th factor contributes $2^{L_{n-1}} 3^{-H_{n-1}}(1 - (2/3)^{m_n}) = 2^{n-1} A_{n-1}(1 - A_n/A_{n-1})$ to $|\zeta_\infty|$, where $A_n = (2/3)^{H_n}$; summing $\sum_n 2^{n-1}(A_{n-1} - A_n)$ by parts gives $A_0 + \frac{1}{2}\sum_n 2^n A_n$. Convergence is geometric of ratio $\frac{1}{2}$ by (2). □

The limit word has ones of density $2\gamma/(2\gamma+1)$, not α, so it is not itself a member of the family; Proposition 17 concerns the limit of the values.

6 A criterion for recurrence

The examples above raise the question of which words of the shape (3) have recurrent trajectories at all. By Lemma 2,

$$3^{D_{n+r} - D_n} = 2^{-(g(n+r) - g(n))} \cdot 3^{\beta_{n+r} - \beta_n}, \qquad \tfrac{2}{3} < 3^{\beta_{n+r} - \beta_n} < \tfrac{3}{2}, \tag{7}$$

splitting the ratio into a deterministic part and a bounded fluctuation. This suggests the scalar

$$R_n = \sum_{r \geq 0} 2^{-(g(n+r) - g(n))}.$$

Theorem 18. *Let f in (3) be nondecreasing with $f(n) \to \infty$, the density being critical, and suppose $\Phi_{\mathbb{R}}(v)$ exists. Then $\frac{2}{9}R_n \leq -z_n \leq \frac{3}{2}R_n$ for all large n, so (z_n) has an accumulation point if and only if $\liminf_n R_n < \infty$. Equivalently, if a run of b_j zeros begins at time t_j then $R_{t_j + b_j} = 2^{-b_j} R_{t_j}$, and recurrence holds iff $\liminf_j 2^{-b_j} R_{t_j} < \infty$.*

The constants come from (7) and from the damping factor of Lemma 3 being eventually at least $\frac{1}{3}$. A finite $\liminf$ gives a bounded infinite subsequence of the z_n and hence a convergent one; if $R_n \to \infty$ then the local minima of R, which occur at the ends of the zero runs, escape, and with them the trajectory.

Corollary 19. *Under the hypotheses of Theorem 18, $\liminf_n R_n < \infty$ implies $\Phi_{\mathbb{R}}(v) \notin \mathbb{Q}_{\text{odd}}$.*

Indeed an infinite subset of $\mathbb{Q}_{\text{odd}}$ has no accumulation point in $\mathbb{R}$ [2, p. 3]. For the word of Theorem 1, $g(n) = \lfloor 5n/6 \rfloor$ gives $R_n \to (1 - 2^{-5/6})^{-1} = 2.2791\ldots$, so §3 is the bounded case of Theorem 18.

Remark 20. Neither obvious candidate governs recurrence. Zero runs tending to infinity do not suffice: taking $g(n) = \lceil c \log(n+1) \rceil$ in (3), so that $f(n) = n - g(n) \sim n$ and the zero runs grow *linearly*, one still has critical density, yet R_n diverges and no accumulation point exists. What matters is whether g outruns $\log_2$, since R_n is bounded exactly when $g(n+r) - g(n) \gtrsim (1+\varepsilon) \log_2 r$. Nor is the drift the invariant: $D_n \to -\infty$ governs convergence of the series, not recurrence, and the word of Theorem 1 and its associated word [2, Example 7] both have $D_n \to -\infty$ with opposite behaviour. By the zero-run form of Theorem 18 the relevant quantity is not the contraction length b_j alone but the scale R_{t_j} on which it acts.

7 Scope

Theorem 1 settles the case left open in [2] and, by [2, p. 3], gives $\zeta \notin \mathbb{Q}_{\text{odd}}$ for that word. It bears on neither half of the Collatz conjecture. Half (C), the absence of nontrivial cycles, is untouched by everything here: cycles correspond to eventually periodic words, and every word considered is aperiodic by construction. Half (D) would follow from the periodicity conjecture, but our results fall short of that for two reasons worth stating precisely.

First, the conjecture quantifies over all aperiodic words, and no uniform mechanism is available here: the associated word of [2, Example 7] has the same critical density and provably no accumulation point, while the method above is parasitic on the structure of (3) — G quadratic and $C_s(r)$ free of M — which a general word does not possess. Theorem 18 identifies when recurrence holds in the class (3), but it yields only the existence of *some* accumulation point, not that ζ is one of them; the latter needs the recurrence of the reduction to a phase at which (6) takes the value ζ, which is what Lemma 8 supplies here and what a general argument would have to replace.

Second, Corollary 19 constrains the real value $\Phi_{\mathbb{R}}(v)$, whereas the conjecture concerns the 2-adic value $\Phi(v)$. These are limits of the same rational partial sums $S_\ell = \Phi_{\mathbb{R}}(u^{(\ell)})$ in different completions and need not agree. The missing implication is that $\Phi(v) \in \mathbb{Q}_{\text{odd}}$ forces $\Phi_{\mathbb{R}}(v) = \Phi(v)$ when the real series converges. It can be located exactly: for a finite prefix there is no ambiguity, $\Phi(u) = \Phi_{\mathbb{R}}(u)$, and combining $T^\ell(x) = (\varphi(u) + 3^h x)/2^\ell$ with $\Phi(u) = -\varphi(u)/3^h$ gives

$$x - S_\ell = \frac{2^\ell}{3^h}\, T^\ell(x) = 3^{\,D_\ell}\, T^\ell(x), \tag{8}$$

so that $\Phi_{\mathbb{R}}(v) = x$ if and only if $T^\ell(x) = o(3^{-D_\ell})$, while $S_\ell \to L \neq x$ forces $T^\ell(x) \sim 3^{-D_\ell}(x - L)$. This is a relocation rather than a weakening: 3^{-D_ℓ} is the natural homogeneous growth scale of the maps being iterated, so the condition carries no slack and is precisely the cancellation selecting $x = \Phi_{\mathbb{R}}(v)$. What (8) provides is a concrete form for what is missing — a growth estimate for the orbit of the hypothetical rational — rather than an abstract incompatibility of completions. We have not been able to supply it.

Finally, Proposition 9 and Table 1 are computer-assisted, and their correctness depends on that of the interval arithmetic; the code is short, uses no third-party library, and was checked against an independent implementation agreeing to twenty significant digits. We have not found the scalar R_n or the equivalence of Theorem 18 in [2], but make no claim of novelty beyond that comparison.

References

[1] D. J. Bernstein and J. C. Lagarias, *The* $3x + 1$ *conjugacy map*, Canad. J. Math. **48** (1996), 1154–1169.

[2] J. López and P. Stoll, *The* $3x + 1$ *periodicity conjecture in* $\mathbb{R}$, arXiv:2101.12747 (2021).

[3] K. G. Monks and J. Yazinski, *The autoconjugacy of the* $3x+1$ *function*, Discrete Math. **275** (2004), 219–236.

[4] L. Kuipers and H. Niederreiter, *Uniform Distribution of Sequences*, Wiley, 1974.